

Birla Institute of Technology and Science, Pilani

Course Project Gas Dynamics

Supersonic and Hypersonic Flow Over Orion Body

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Abstract

Understanding various aerothermal loads experienced by a re-entry capsule helps us ensure the safe return of astronauts and the various experimental data that they have collected. Predicting the effects of these aerothermal loads in various flow regimes, supersonic or hypersonic, is a challenge taken up by research institutes around the world. This work attempts to model the same on an Orion Body, simulating the flow conditions in ANSYS using a k-ϵ Turbulence Model while considering Sutherland Viscosity Law and verifies its findings with its predecessors.

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1 Introduction

Humans are now in the process of exploring the final frontier, space. Sending people to space is a monumental task, getting them back safely is also equally arduous. Atmospheric re-entry is a challenge for manned space missions. It is of paramount importance that one correctly models the various aerothermal loads experienced by the re-entry vehicle to ensure the safety of the crew. For this course project NASA's Orion Crew Exploration Vehicle (CEV), belonging to NASA's now cancelled Constellation Program, has been considered. The Orion CEV uses the same basic configuration as the Apollo Command and Service Module (CSM) that was used during the first manned journey to the moon.

An extensive literature review was conducted in preparation for this project and it was found that several experimental and computational tests have been conducted across the world in various universities to dedicated research facilities. For example, in the subsonic regime, [1] computed the behaviour of the flow field at subsonic Mach Numbers using URANS (Unsteady Reynolds-Averaged Navier-Stokes) . [2] performed wind tunnel and computational studies to develop the static aerodynamic database for the Orion CEV. [3] studied the effect of geometrical parameters of the re-entry capsules on the flow field characteristics, using various capsules such as ARD, Soyuz and OREX, while using time-dependent, axisymmetric, compressible laminar Navier-Stokes equations. [8] also analysed the aerodynamics of the Orion CEV at high altitudes using DSMC method to solve the Boltzmann equation of kinetic theory and the Reynolds-Average Navier-Stokes methods

along with Sparat-Allmaras turbulent model to compute flow field at low altitude. [4] also conducted and reviewed the comprehensive investigations on the dynamic stability of blunt body atmospheric entry vehicles, paying attention to instabilities such as unstable pitching motions and divergence of oscillation amplitude. For this project two flows were considered, one in the supersonic regime and the other in the hypersonic regime. The detailed specifications of the problem are illustrated in Section 2. The geometric constraints considered for the model have been taken from the paper [5] and a detailed schematic of the same has been included in Section 4. The entirety of the model was simulated in ANSYS with certain governing equations and other constraints given in Section 3 of this report. The axisymmetric model was created in SpaceClaim and meshed as per requirements. The turbulent flow analysis was conducted using a Realizable k-ϵ Turbulence Model and the Sutherland Viscosity Law was used to model the regions where viscous effects dominated the flow characteristics. Detailed information about the same is given in Sections 5 to 7. This work aims to highlight the variations in the Pressure, Temperature, Velocity, Mach Number and the coefficient of Pressure over the Orion CEV when subjected to supersonic and hypersonic flows. The results given in the paper [6] were used to verify and validate the results obtained which have been explained in Section 8 of this report. Other important results have been included in Section 9. This report also analyses the characteristics of Hypersonic Flows.

2 Problem Specification

A uniform supersonic and later, hypersonic stream encounters the orion body at an angle of attack (α) of 0 degrees as shown in Figure 1.

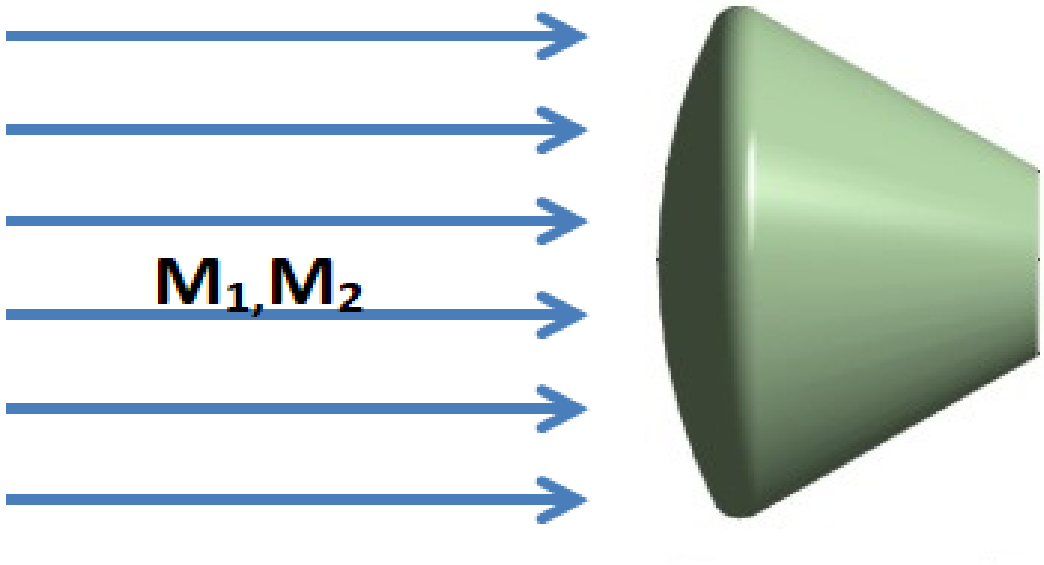


Figure 1: Schematic Diagram of the Problem

The stream is at the following conditions:
Case 1:

- Mach Number $M_1 = 2$
- Freestream Pressure $p_1 = 2891$ Pa
- Freestream Temperature $T_1 = 219$ K

Case 2:

- Mach Number $M_2 = 6$
- Freestream Pressure $p_2 = 1064$ Pa
- Freestream Temperature $T_2 = 234$ K

3 Pre-Analysis

The compressible flow solutions generally require solving the energy equation, compressibility effects couple hydrodynamics and thermodynamics necessitating the simultaneous solution of the continuity, momentum and energy equations. There is an extra dependence of density on pressure and temperature in compressible flows as compared to incompressible gas flows.
The pressure-based approach based on the SIMPLE-family algorithm is usually preferred in Mach number spectrum problems. The reason being the pressure-based approach can resolve fluid flows in the various Mach number regimes without any artificial treatment to promote convergence and stabilise computations. Focusing on supersonic flows, changes in velocity become relatively small as compared to the magnitude of the velocity, indicating that variations in pressure do significantly affect density. Hence pressure now acts on density alone through the equation of state to satisfy mass conservation, and the continuity equation can be viewed as the equation of density. In this report the Pre-Analysis was done using [9].

3.1 The conservation Equations

The conservation equations for solving compressible flows include the continuity, momentum and energy equation equations. For Newtonian fluids, the equations are stated below:

Continuity Equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho v) = 0 \tag{1}$$

Momentum Equation:

$$\frac{\partial [\rho v]}{\partial t} + \nabla \cdot \rho v v = \nabla \cdot \mu \nabla v - \nabla p + \mu (\nabla v)^T - \frac{2 \cdot \nabla (\mu \nabla \cdot v)}{3} + f_b \tag{2}$$

Energy Equation:

$$\frac{\partial (\rho C_p T)}{\partial t} + \nabla \cdot [\rho C_p v T] = \nabla \cdot [k \nabla T] + \rho T \frac{D C_p}{D t} + \frac{D p}{D t} - \frac{2 \mu \psi}{3} + \mu \Phi + \dot{q} v \tag{3}$$

The above equations should be appended by an equation of state relating density to temperature and pressure:$\rho = (p, T)$. The solver can't directly take into considerations these three equations as they are continuous equations and the solver itself calculates on discrete points. Hence discretisation of each of the three equations is necessary. This is done by modifying the equation accordingly.

- **The discretisation of the Momentum Equation:** Discretisation based on the pressure correction equation.
- **The discretisation of the Energy Equation:** The discretisation of the unsteady, convection and diffusion terms of the energy equation is done by considering the additional terms. Namely, specific heat term, substantial derivative term, viscous dissipation term, source/sink term.
- **The discretisation of the Continuity Equation:** It is done by integrating over the control surface and using divergence theorem.

3.2 Boundary Conditions

A boundary element has at least one face located at a boundary patch, which is denoted as a boundary face. The treatment of boundary conditions at a boundary face is critical to the accuracy and robustness of any CFD code. The boundary conditions for these three equations are fed by the solver using the overall inlet and outlet boundary conditions we feed in. There are differential mathematical techniques based on solving those three equations with the help of wall (inlet and outlet) and symmetry boundary conditions.The inlet boundary conditions have three types. (i) specified velocity;(ii) specified static pressure and velocity direction; and (iii) specified total pressure and velocity direction. The outlet boundary conditions have three types. (i) a specified static pressure, (ii) a specified mass flow rate, and (iii) a fully developed flow. In symmetry boundary conditions, for scalars, they are simply reflected by setting its normal gradient to be zero. For vectors, it is the same but with the parallel component to retaining magnitude and direction.There are inlet and outlet boundary conditions for the flow domain which is given by the user. The boundary conditions for Supersonic and Hypersonic flow are:

- **Inlet Boundary Conditions:**
  Specified static pressure, velocity, and temperature (Pb = Pspecified ; vb = Vspecified ; T = Tspecified)
  This is equivalent to Dirchilet Type condition, and thus the coefficient of the boundary cell is found to be
  $$a_C^p = \frac{V_c C_p}{\Delta t} + \sum_{f \ nb(c)} \left(\frac{C_{\rho,f} \parallel \dot{m}_f^*, 0 \parallel}{\rho_f^*}\right) + \sum_{f \ nb(c)} \rho_f^* D_f \tag{4}$$
- **Outlet Boundary Conditions:** At a supersonic outlet none of the variables should be specified and the values of pressure, velocity, density, and temperature are extrapolated from the interior of the Domain. This is equivalent to Neuman boundary conditions, and hence the coefficient of the boundary cell is found to be
  $$a_C^p = \frac{V_c C_p}{\Delta t} + \sum_{f \ nb(c)} \left(\frac{C_{\rho,f} \parallel \dot{m}_f^*, 0 \parallel}{\rho_f^*}\right) + \sum_{f \ nb(c)} \rho_f^* D_f + \left(\frac{\dot{m}_b^*}{\rho_b^*}\right) C_{p,b} \tag{5}$$

3.3 Turbulence Modelling

Non-turbulent flows can be modelled through the Navier-Stokes equation. In contrast, turbulent flow is chaotic, diffusive causing rapid mixing, time-dependent, and involve three dimensional vorticity fluctuations with a broad range of time and length scales. Most accepted theory on turbulence is based on the energy cascade concept.

Based on the energy cascade concept a direct numerical solution of the Navier-Stokes equation for turbulent flows necessitates the use of a tiny time step and fine mesh resulting in a large number of grid points and hence considerable computational time is required to solve using the Direct Numerical Solution(DNS). To save the computational time, a statistical time-average solving model is used called the Reynolds-Average Navier-Stokes Equations.

In Reynolds-average Navier-stokes (RANS) equation of motion for fluid flow is time-averaged. The idea behind the equation is Reynolds decomposition, whereby an instantaneous quantity is decomposed into its time-averaged and fluctuating portions. Then RANS equations are primarily used to describe turbulent flows. These equations can be used with an approximation based on knowledge of the properties of flow turbulence to give approximate time-averaged solution to the Navier-Stokes equations. The Navier-Stokes momentum equation in the convective form is given below:

$$\rho\frac{Du}{Dt} = \rho(\frac{\partial u}{\partial t} + u \cdot \nabla u) = -\nabla\tilde{p} + \mu\nabla^2 u + \frac{\mu\nabla(\nabla \cdot u)}{3} + \rho g \tag{6}$$

The Reynolds-average Navier-Stokes equation (RANS) is given below:

$$\rho\tilde{u_j}\frac{\partial\tilde{u_i}}{\partial x_j} = \rho\tilde{f_i} + \frac{\partial[-\tilde{p}\delta_{ij} + \mu(\frac{\partial\tilde{u_i}}{\partial x_j} + \frac{\partial\tilde{u_j}}{\partial x_i}) - \rho u_i\tilde{u}_j]}{\partial x_j} \tag{7}$$

Several turbulence models are being developed to express the turbulent viscosity in terms of velocity and length.

$$\mu = \sigma \cdot l \cdot \sqrt{k} \tag{8}$$

These models are grouped into four main categories:

- Algebraic (zero-equation) model
- One-equation model
- Two-equation model
- Second-order closure model

Algebraic models are used to compute μ without solving differential equations. The one-equation models require solving only one transport differential equation to calculate the turbulence data. Two-equation turbulence model necessitates the solution of two transport equation to compute μ . The second-order closure model is computationally costly, so it is not used in this work. The two-equation turbulence models are the most popular in terms of usage in the simulation of industrial applications, requiring the solution of two transport equations while delivering accurate enough predictions. In this report the two-equation model is used. There are various models itself in the two-equation model, but the k-ϵ model has been used.

3.4 The two-equation k-ϵ model

The k-ϵ model is based on the Boussinesq approximation with the turbulent viscosity μ and thermal diffusivity k formulated as

$$\mu = \frac{\rho C_\mu k^2}{\epsilon} \tag{9}$$

$$k = \frac{C_p \mu}{Pr} \tag{10}$$

where ϵ is the rate of dissipation of turbulence kinetic energy per unit mass due to viscous stresses.

$$\epsilon = \mu \frac{(\nabla v' + (\nabla v')^T) \vdots (\nabla v' + (\nabla v')^T}{2\rho} \tag{11}$$

The corresponding equations are given below:

$$\frac{\partial(\rho k)}{\partial t} + \nabla \cdot (\rho v k) = \nabla \cdot (\mu_{eff,k} \nabla k) + P_k - \rho\epsilon \tag{12}$$

$$\frac{\partial(\rho \epsilon)}{\partial t} + \nabla \cdot (\rho v \epsilon) = \nabla \cdot (\mu_{eff,\epsilon} \nabla \epsilon) + C_{\epsilon 1} \frac{\epsilon}{k} P_k - C_{\epsilon 2\rho} \frac{\epsilon^2}{k} \tag{13}$$

where

$$\mu_{eff,k} = \mu + \frac{\mu_t}{\sigma_k}, \mu_{eff,\epsilon} = \mu + \frac{\mu_t}{\sigma_\epsilon} \tag{14}$$

The standard $k - \epsilon$ model is a high Reynolds number turbulence model valid only for fully turbulent free shear flows that cannot be integrated to the wall. Based on this two-equation $k-\epsilon$ model, a modified recently developed realizable $k - \epsilon$ model is developed. They both differ by:

- The realizable $k - \epsilon$ model contains a new formulation for the turbulent viscosity.
- A new transport equation for the dissipation rate, ϵ, has been derived from an exact equation for the transport of the mean-square vorticity fluctuation.

An immediate benefit of the realizable k-ϵmodel is that it more accurately predicts the spreading rate of both planar and round jets. It is also likely to provide superior performance for flows involving rotation, boundary layers under strong adverse pressure gradients, separation, and recirculation.

The modelled transport equation for the realizable k-ϵmodel are:

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial(\rho k u_{ij})}{\partial x_j} = \frac{\partial[(\mu + \frac{\mu_t}{\sigma_k})\frac{\partial k}{\partial x_j}]}{\partial x_j} + G_k + G_b - \rho\epsilon - Y_M + S_k \tag{15}$$

$$\frac{\partial(\rho \epsilon)}{\partial t} + \frac{\partial(\rho \epsilon u_{ij})}{\partial x_j} = \frac{\partial[(\mu + \frac{\mu_t}{\sigma_\epsilon})\frac{\partial \epsilon}{\partial x_j}]}{\partial x_j} + \rho C_1 S_\epsilon - \rho C_2 \frac{\epsilon^2}{k + \sqrt{v\epsilon}} + C_{1\epsilon} \frac{\epsilon}{k} C_{3\epsilon} G_b + S_\epsilon \tag{16}$$

where

$$C_1 = max[0.43, \frac{\eta}{\eta+5}], \eta = S\frac{k}{\epsilon}, S = \sqrt{2S_{ij}S_{ij}} \tag{17}$$

$$C_{1\epsilon} = 1.44, C_2 = 1.9, \sigma_k = 1.0, \sigma_\epsilon = 1.2 \tag{18}$$

Seeing the advantages stated above, two-equation realizable k-ϵ model is used in ANSYS Fluent Solver.

3.5 Viscosity, Specific Heat Capacity and Thermal Conductivity model

The models described in this sub section are adopted from [6].

- **Sutherland's Viscosity Law:**
  Viscosity is defined by Sutherland's law of viscosity, based on the kinetic theory of gases and idealised intermolecular-force potential.

$$\mu = \mu_o + [\frac{T}{T_o}]^{1.5}(\frac{T_o + Su}{T + Su}) \tag{19}$$

  where $\mu_o$is the reference viscosity, T_o is the reference temperature and Su is the Sutherland constant.

- **Specific Heat Capacity:**
  The piecewise polynomial equation defines the variation of specific heat capacity, C_p as a function of temperature in the range of 550–2000 K. C_p is taken to be 1018.2 J/(kg K) for the temperature range 273–550 K.

$$C_p(T) = a_o + a_1T + a_2T^2 + a_3T^3 + a_4T^4 \tag{20}$$

a_o	a_1	a_2	a_3	a_4
874.687	0.325431	$-2.07132 * 10^{-5}$	$-6.63866 * 10^{-8}$	$2.66353 * 10^{-11}$

Table 1: Heat Capacity Constants

- **Thermal conductivity:**
  Thermal conductivity k(T) variation is calculated from the following piecewise-linear equation at 36 data points of temperature from 250 K to 2000 K in steps of 50 K:

$$k(T) = \frac{1.992 \times 10^{-3} \times T^{1.5}}{T + 112} \tag{21}$$

4 Geometry

This report analyses supersonic flow and hypersonic flow around the Orion body. The body experiences this type of flow during vehicle re-entry into the atmosphere. The original spacecraft is shown in the Figure 2.



Figure 2: Orion CEV

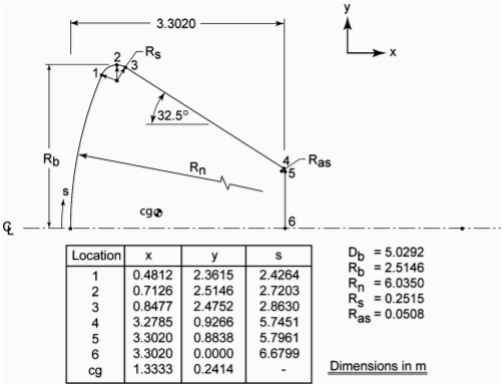


Figure 3: Orion Reference Geometry

The spacecraft body has an axis of rotation, and the spacecraft is symmetric about that axis too. Hence the spacecraft was modelled as a 2-D axis symmetric case. The geometry was adopted from [5] and is given in Figure 3

Using SpaceClaim, a solid modelling CAD software integrated with ANSYS; a 2-D model of the spacecraft was modelled with the dimensions given in the reference geometry in Figure 3. Since the outside effects of the spacecraft have to be analysed, a negative geometry of the body was modelled. The distance between the inlet boundary and the body is kept 3m and between the outlet boundary and the body is 5.7m. This is to observe the compressible flow before and after the air flows over the body. This geometry would be used for further analysis and simulation of the compressible flow across it. The SpaceClaim geometry model is given in Figure 4

5 Mesh

Meshing is an integral part of the engineering simulation process where complex geometries are divided into simple elements that can be used as discrete local approximations of the larger domain. The mesh influences the accuracy, convergence and speed of the simulation.

The geometry is imported from Spaceclaim to the ANSYS Meshing. A mesh has various properties which can be set according to the requirements. Face meshing is used to specify which face of the body needs to mesh and accordingly change the parameters of that particular mesh. Body Sizing is used to alter the parameters related to the size of the mesh element. Edge sizing is used to specify the number of elements on

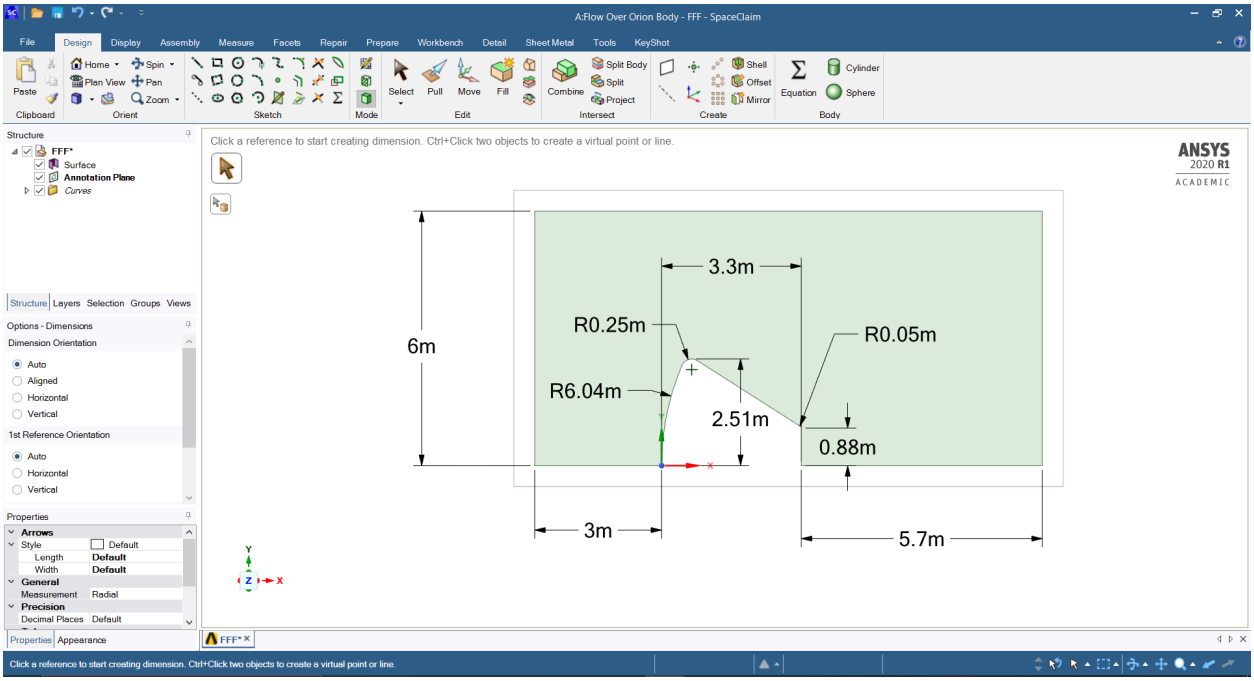


Figure 4: SpaceClaim Model

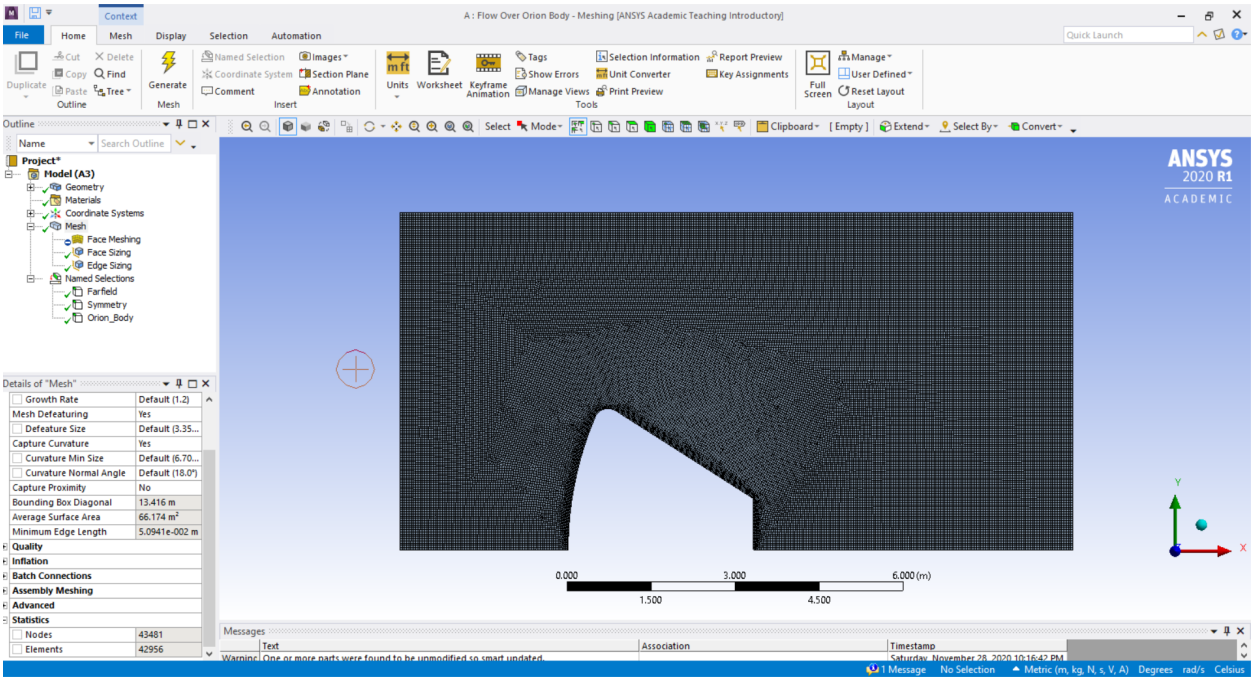


Figure 5: Meshing

a particular edge of the body. This function is beneficial for edges having boundary layers and would need more mesh elements for better results.
The geometry first meshes with the default settings. Then the body sizing parameter is added. The element size is fed as 0.004m with advanced behaviour as soft, which would keep the element size close to the given value if they are constraints and not exact which could cause problems to other parameters. Furthermore, the Edge sizing parameter is added. The whole edge containing the Orion body is selected, and the element size is fed as 0.002m under advanced soft behaviour. This means that the entire edge is divided such that each mesh touching that edge would have a length of 0.002m, which is less than the body size length hence ensuring more mesh elements near the Orion body which is required for better results as there is Fluid Boundary Layer in those meshes.
Finally, the mesh is set to be computed under the above constraints by the solver. The solver preference is set as CFD FLUENT. The final solved mesh parameters are given in the Table 2. All the above-given parameters are reflected in the solved mesh with minor fluctuations because of the advanced soft behaviour. The final solved mesh in the ANSYS Mesh solver is given in the Figure 5.

Minimum edge length	0.00509m
Average Surface Area	$66.174m^2$
Nodes	43481
Elements	42956

Table 2: Mesh Properties

- **Named Selection:**
  Edges are named for further reference while configuring the physical setup of the simulation. Three named selection are created, namely Farfield, Orion Body and Symmetry. The Farfield is the outer boundary of the whole body where inlet and outlet conditions would be set. The Orion body selection is the whole Orion body curve. The Symmetry selection is the entire edge which acts as the axis of symmetry and rotation of the Orion body. All three named selections are saved in the solver along with the created mesh. The named selection are as shown in Figure 6.

6 Physics Setup

6.1 Initial settings

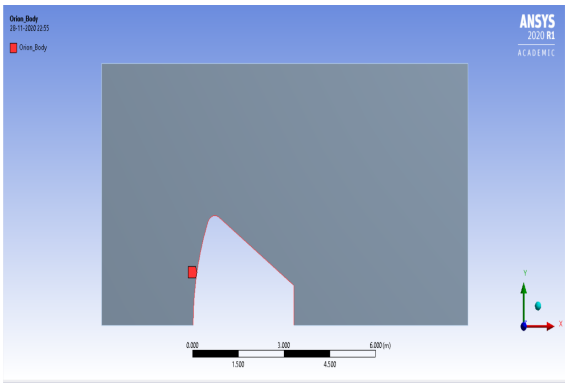
Double click setup in workbench project page, fluent launcher appears. Double precision should be used in order to increase not only precision but also range of magnitudes that can be obtained. But the disadvantage of double precision is it requires more memory.

6.2 Problem setup- General settings

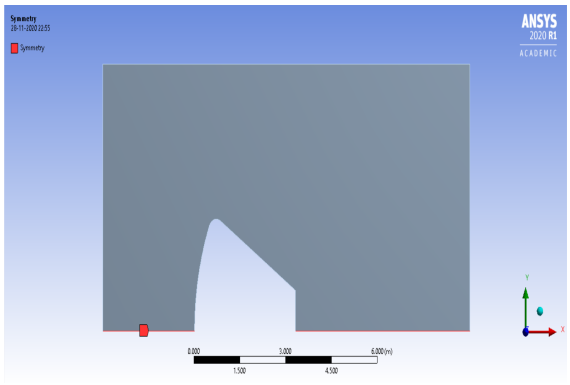
From the outline window click General. Density-based solver should be used.

6.3 Models

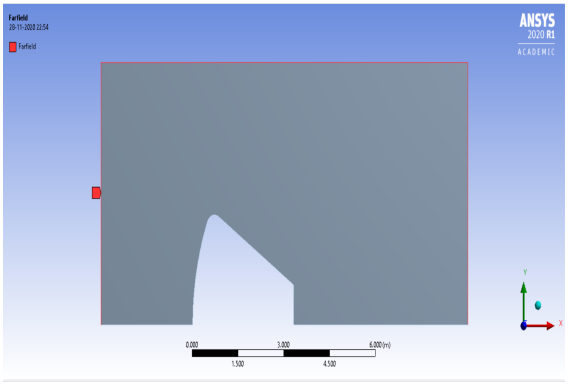
From the outline window click models. Utilize the energy equation in order to solve this simulation. Under Models highlight Energy-off and click Edit Now, the Energy window will launch. Check the box next to



(a) Orion Body Named Selection



(b) Symmetry Named Selection



(c) Farfield Named Selection

Figure 6: Named Selections

Energy Equation and hit OK. Doing this turns on the energy equation. Change the type of viscosity model. Select Viscous - Laminar and click Edit, Choose the k-epsilon option (which is the most used CFD turbulence model) and make sure that realizable k-epsilon is the model which is chosen and press OK.

6.4 Materials

In the Outline window, highlight Materials. In the Materials window, highlight Fluid, and click Create or Edit this will launch the Create or Edit Materials window; here specify the properties of the fluid. Set the Density to Ideal Gas, and the Molecular Weight (28.966) are used. Piecewise-polynomial(fourth degree) for specific heat and the coefficients are as shown in table 1 have been used. Thermal conductivity k variation is calculated from the piecewise-linear equation 20 at 36 data points of temperature from 250 K to 2000 K in steps of 50 K. Sutherland is chosen for viscosity. Update these fields and press Change/Create.

6.5 Boundary Conditions

In the Outline window, select Boundary Conditions. Now specify each boundary condition for the simulation.

- **Farfield**
  In the Boundary Conditions window, select farfield. Use the drop-down menu to change the Type to pressure-far-field.To confirm the change, press OK. Next, a dialogue box will open with some parameters to be specified. Change the Gauge Pressure (Pascal) to 2891, and Mach Number to 2 and the

temperature to 219K.After finishing, press OK.

- **Body**
  In the Boundary Conditions window, select Orion-body. Use the drop-down menu to change the Type to wall.

- **Symmetry**
  In the Boundary Conditions window, select symmetry. Use the drop-down menu to change the Type to symmetry.

- **Operating Conditions**
  In the Boundary Conditions window, select the Operating Conditions button. Change the Gauge Pressure to 0.

6.6 Reference values

Enter the reference values as shown in Figure 7

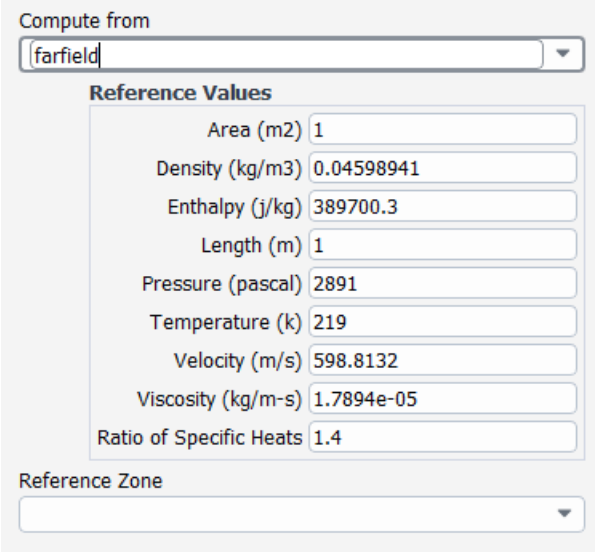


Figure 7: reference values

7 Numerical Solution

7.1 Solution Methods

Select Solution Methods to open the Solution Methods window. Under Spatial Discretization, ensure that the option under Flow Second Order Upwind is selected.

7.2 Solution Controls

Select Solution Controls to open the Solution Controls window. Ensure that the Courant Number is set appropriately.

7.3 Monitor

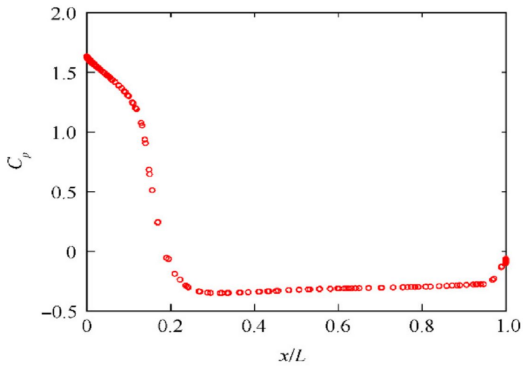
Open the Monitors window. In the Monitors window, select Residuals - Print, Plot and press Edit. This will open the Residual Monitors window. Convergence criteria should be changed for the given problem. Under Equation and to the right of Continuity, change the Absolute Criteria to 10^{-6}. Repeat for x-velocity, y-velocity, and energy, then press OK.

7.4 Initialization & Calculation

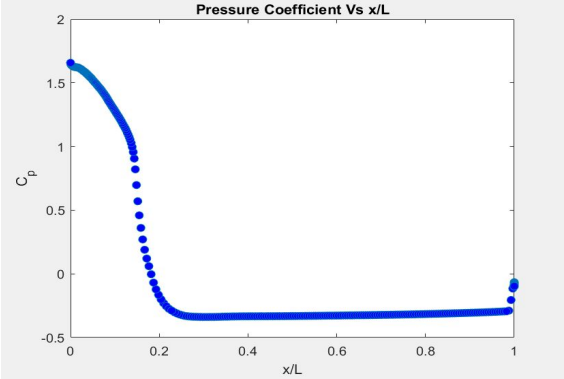
Select Solution Initialization. In this window, select Standard Initialization. Check to see that the values that generate match our inputted farfield values, then press Initialize. In the Outline window, tick solution steering and choose the appropriate flow condition. The number of iteration for the supersonic case was 20 and 1000 for the hypersonic case. Double click Calculate to run the calculation.

8 Verification and Validation

In [6], the plots of pressure coefficient vs axial position plotted along the sectional surface of baseline re-entry vehicle for various Mach numbers were obtained by the authors. The plots for C_p Vs x/L for Mach 2 and Mach 6 obtained in this report are compared with the corresponding plots in [6].



(a) Plots for Mach 2 as obtained in (Reference)



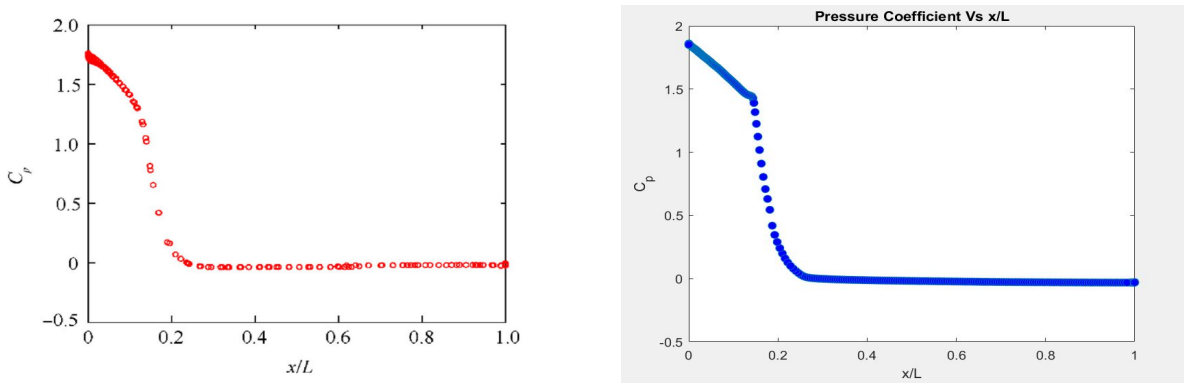
(b) Simulation Result for Mach 2

Figure 8: C_p Vs x/L for Mach 2

Figure 8 and Figure 9 shows the plots of pressure coefficient (C_p) on the surface of Orion body Vs the ratio of axial distance and axial length of Orion body (x/L) for Mach 2 and Mach 6 respectively. It is clear from the figure that the simulation is matching with the results obtained in [6]. The pressure coefficient is a non-dimensional number and is given as:

$$C_p = \frac{p - p_\infty}{0.5\rho_\infty V_\infty^2} \tag{22}$$

where, p is static pressure at the point.
p_∞ is the static pressure at freestream.
ρ_∞ is the freestream fluid density.
V_∞ is the freestream velocity of fluid.
The dimensionless ratio and C_p values at various points (Refer Figure 3) are:



(a) Plots for Mach 6 as obtained in (Reference)

(b) Simulation Result for Mach 6

Figure 9: C_p Vs x/L for Mach 6

Table 3: Non-dimensional distance and C_p for various points

Points	x	x/L	$C_p(M=2)$	$C_p(M=6)$
Tip	0	0	1.657	1.85
1	0.481	0.146	0.8209	1.39
2	0.713	0.216	-0.2541	0.1602
3	0.848	0.257	-0.3294	0.0199
4	3.278	0.993	-0.2056	-0.0318
5	3.302	1	-0.0662	-0.0269

For the Supersonic case (M = 2) it is clear from Figure 8b that the pressure at the Tip of Orion Body is highest because of formation of bow shock. The pressure decreases from Tip to Point 1 due to the shape of Orion body and there is a drastic decrease in pressure at Point 2 because of formation expansion wave. The expansion wave starts forming after Point 1 and ends at Point 3 due to which the pressure decreases drastically. From Point 3 to Point 4 the pressure is almost constant. Pressure starts to increase slightly from Point 4 to Point 5 because of the formation of weak oblique shock.

A similar trend is observed for the Hypersonic Case (M = 6) as shown in Figure 9b but the magnitude of pressure are higher from Tip to Point 1. The pressure then reduces from Point 1 to Point 3 due to the presence of expansion wave. The pressure strength of the weak oblique shock between Point 4 and Point 5 is:

$$\left.\frac{\Delta P}{P}\right|_{M=2} = 0.68$$

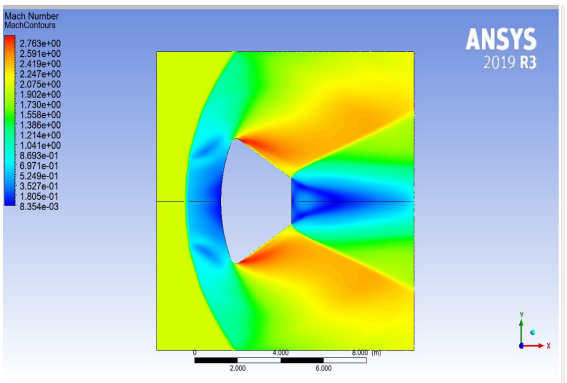
$$\left.\frac{\Delta P}{P}\right|_{M=6} = 0.154$$

Hence, for the supersonic case a stronger oblique shock is observed compared to hypersonic case.

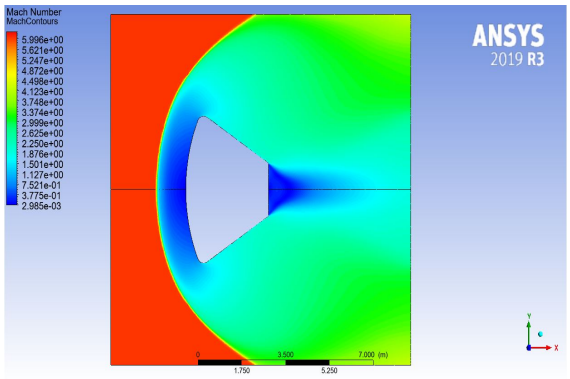
9 Results and Discussion

9.1 Mach Number

The location of shocks can be easily identified with the plots of Mach number. Figure 10 shows the Mach Contours for Supersonic flow (Figure 10a) and Hypersonic flow (Figure 10b).



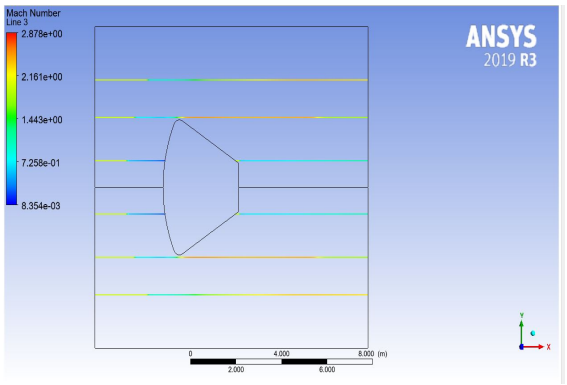
(a) Mach 2 contours



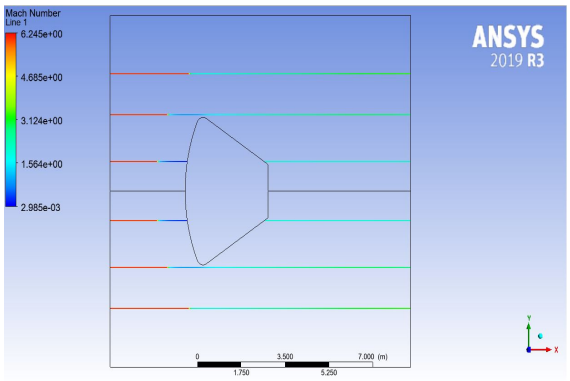
(b) Mach 6 contours

Figure 10: Mach Contours

To better understand the location of shock waves the Mach number was plotted along x for various values of y. Figure 11 shows lines for which Mach number was plotted.



(a) Location of lines for Supersonic Case



(b) Location of lines for Hypersonic Case

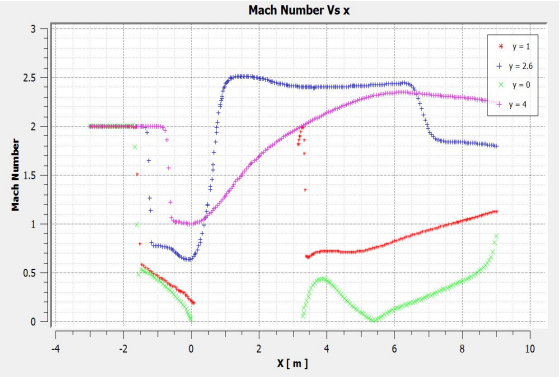
Figure 11: Location of Lines

Figure 12 shows the variation of Mach Number along x for various y values. For the Supersonic Case (Figure 12a) the following observation can be made:

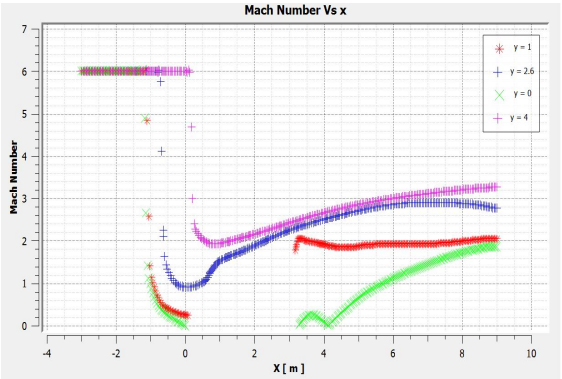
- For y = 0 i.e symmetry line the bow shock is observed at x = -1.75 m which decreases the Mach number from 2 to 0.5. Subsequently as x increases the Mach Number decreases as the fluid comes closer to the Orion body (becoming 0 at the tip). Just behind the Orion body (x > 3.3 m) the Mach Number first increases and becomes maximum at x = 3.75 m and then decreases and become 0 at x = 5.4 m and then starts increasing for x > 5.4 m.

- For y = 1 m above symmetric line the bow shock is observed at x = - 1.8 m due to which the Mach number decreases from 2 to 0.3 (the flow turns over the Orion body). The increase in Mach Number at x = 3.3 is due to the presence of expansion wave (the effects are very small). Due to oblique shock the mach number then decreases and then starts increasing finally reaching Supersonic flow.

- For y = 2.6 m above the symmetric line the box shock is observed at x = -1.45 m which decreases the Mach Number from 2 to 0.8.The effect of expansion wave are clearly visible from x > 0 to x = 1 m



(a) Mach Number Vs x for Supersonic case



(b) Mach Number Vs x for Hypersonic case

Figure 12: Mach Number Vs x

which increase the Mach Number from 0.7 to 2.5. The oblique shock is encountered at x = 6.5 m due to which the Mach number decreases from 2.4 to 1.8.

- For y = 4 m above the symmetric line the bow shock is observed at x = -1 m which decreases the Mach Number from 2 to 1. The effects of expansion wave are significant and it is observed from x = 0 to x = 6 increasing the Mach Number from 1 to 2.6. A very weak oblique shock is observed at x = 6.

For the Hypersonic Case (Figure 12b) the following observation can be made:

- For y = 0 i.e symmetry line the bow shock is observed at x = -1.25 m which decreases the Mach number from 6 to 1. The bow shock is more closer to the Orion body which is a direct consequence of hypersonic flow. Subsequently as x increases the Mach Number decreases as the fluid comes closer to the Orion body (becoming 0 at the tip). Just behind the Orion body (x > 3.3 m) the Mach Number first increases and becomes maximum at x = 3.75 m and then decreases and become 0 at x = 4.2 m and then starts increasing for x > 4.2 m.

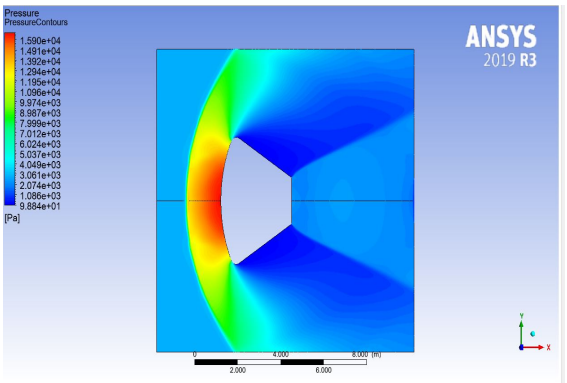
- For y = 1 m above symmetric line the bow shock is observed at x = -1.25 m due to which the Mach number decreases from 6 to 1.1 (the flow turns over the Orion body). The increase in Mach Number from x = 3.3 to x = 3.5 is due to the presence of expansion wave. A very weak oblique shock is observed due to which the mach number decreases and then starts increasing.

- For y = 2.6 m above the symmetric line the bow shock is observed at x = -0.8 m which decreases the Mach Number from 6 to 1.The effect of expansion wave are clearly visible from x > 0 to x = 7 m which increase the Mach Number from 1 to 2.9. The oblique shock is encountered at x = 7.5 m due to which the Mach number decreases very slightly.

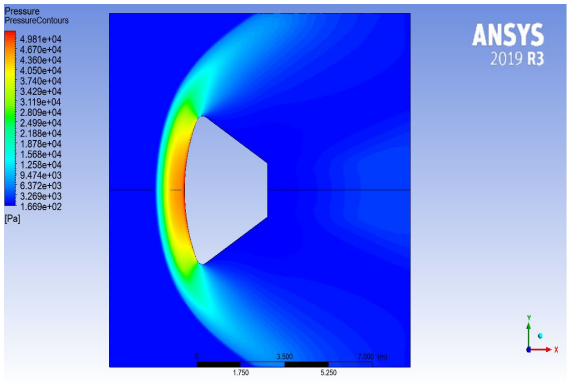
- For y = 4 m above the symmetric line the bow shock is observed at x = 0.1 m which decreases the Mach Number from 6 to 2. The effects of expansion wave are significant and it is observed for x ¿ 0 increasing the Mach Number from 2 to 3.1.

9.2 Pressure

Figure 13 clearly shows the pressure contours and it can be observed that the bow shock is much more closer to the Orion body in the Hypersonic case then in the Supersonic case.



(a) Pressure contour for Supersonic case

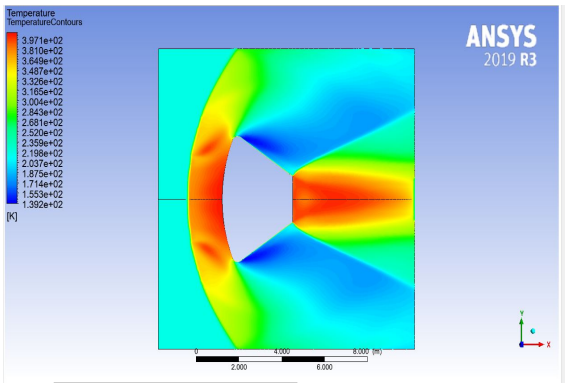


(b) Pressure contour for Hypersonic case

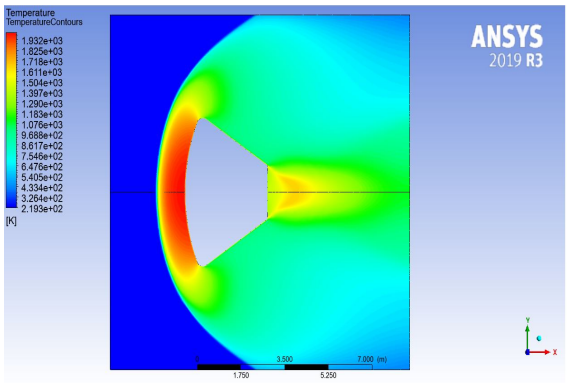
Figure 13: Pressure contour

9.3 Temperature Contour

Figure 14 shows the Temperature contours and the effects of bow shock, expansion wave and oblique shock are clearly visible in the plots. The Temperature increases through a bow shock, it then decreases through the expansion wave and increases again through the oblique shock.



(a) Temperature contour for Supersonic case



(b) Temperature contour for Hypersonic case

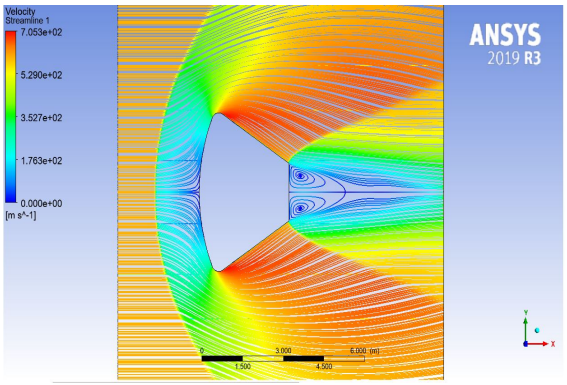
Figure 14: Temperature contour

9.4 Velocity

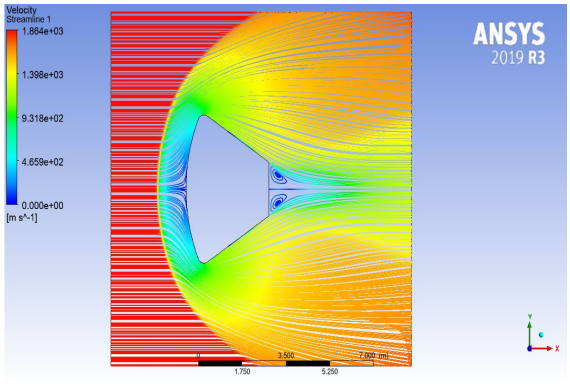
Figure 15 shows the velocity streamlines for the Supersonic case (Figure 15a) and Hypersonic case (Figure 15b). The bow shock, expansion wave and oblique shock are clearly visible in the velocity streamlines. Moreover, a vortex is formed behind the Orion body for both the cases. A major difference between Supersonic and Hypersonic flow is that all the shocks and vortices for the Hypersonic case are closer to the Orion body then the Supersonic case.

9.5 Entropy

Figure 16 shows the entropy contours for both Supersonic (Figure 16a) and Hypersonic case (Figure 16b). The increase in entropy behind the Orion body is much more pronounced and closer to the Orion body in the



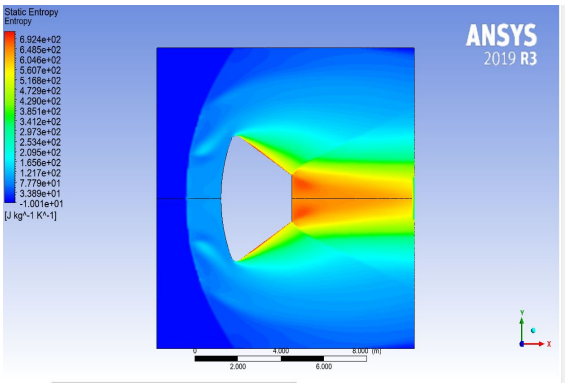
(a) Velocity Streamlines for Supersonic case



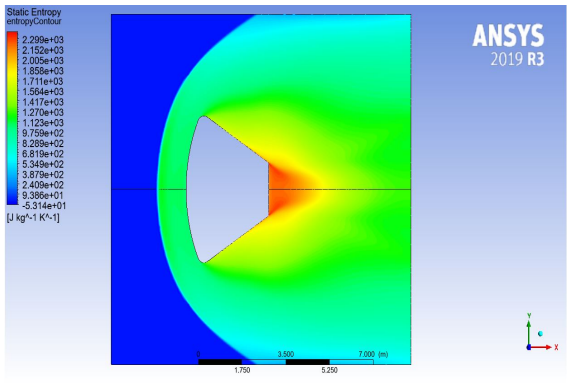
(b) Velocity Streamlines for Hypersonic case

Figure 15: Velocity Streamlines

Hypersonic case then in Supersonic case. Moreover the entropy boundary layer between Point 2 and Point 3 (Refer Figure 3) in the case of Hypersonic flow is much higher in magnitude as compared to Supersonic case, which is another characteristic of Hypersonic flow.



(a) Entropy Contours for Supersonic case

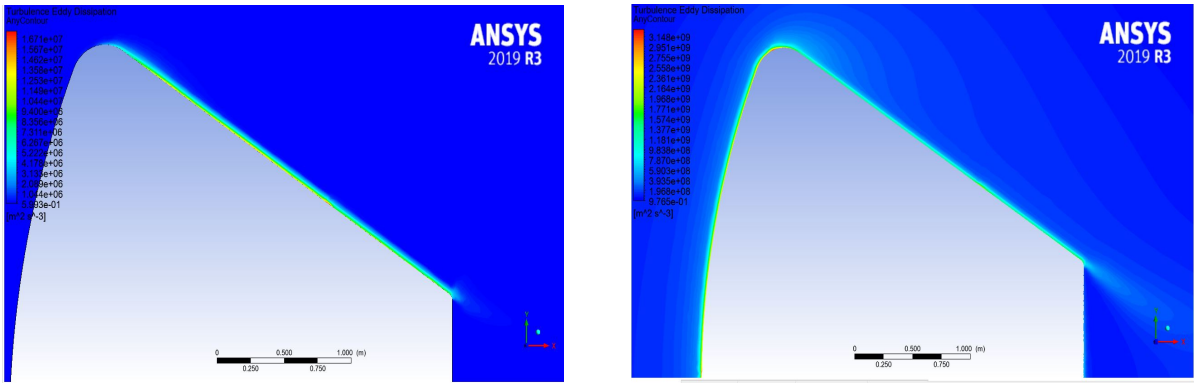


(b) Entropy Contours for Hypersonic case

Figure 16: Entropy Contours

9.6 Turbulent Eddy Dissipation

Figure 17 shows the Turbulent Eddy Dissipation contours for both Supersonic (Figure 17a) and Hypersonic case (Figure 17a). The magnitude of dissipation is much larger in case of Hypersonic flow then in Supersonic flow.



(a) Turbulent Eddy Dissipation for Supersonic case

(b) Turbulent Eddy Dissipation for Hypersonic case

Figure 17: Turbulent Eddy Dissipation

10 Conclusion

In this report the Supersonic flow (Mach 2) and Hypersonic flow (Mach 6) over the Orion body was analysed. The simulation was conducted in ANSYS and the detailed procedure was also explained in this report. The simulation for both the flow scenario was first verified with existing literature and then both the flows were analysed in detail. In the analysis it was observed that the bow shock, expansion wave, oblique shock and vortexes were much closer to the Orion body in the case of Hypersonic case as compared to Supersonic case. The entropy layer and turbulence eddy dissipation in case of Hypersonic case was much more pronounced. Many of the defining characteristics for Hypersonic flow were observed in this report. The future work includes the modelling of chemical reactions due to high temperature and rarefied gas dynamics.

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